Robust PSD estimation for LISA

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1 Introduction

The Power Spectral Density (PSD) characterizes the power of a time series over different frequencies. In Laser Interferometer Space Antenna (LISA), transient bursts – astrophysical signals or non-Gaussian noise artifacts known as glitches – can significantly alter the PSD estimate. We review `mean-based` PSD estimation methods described in [1] and examine their robustness to transients. To address their limitations, we introduce a new `median-based` method for PSD estimation and derive the probability distribution for these estimators.

2 `Mean-based` PSD estimation methods

2.1 WOSA method

Welch's Overlapped Segment Averaging ([1], [2]) is a practical approach for estimating the PSD of a time series. The main steps are as follows.

1. Divide the time-series into possibly overlapping segments.
2. Apply a window function (such as a Hann window) to each segment to reduce spectral leakage.
3. Compute the discrete Fourier transform (DFT) of each windowed segment and normalize by the window's power.
4. Average the normalized periodograms across all segments to reduce the variance and obtain the final PSD estimate.

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2.2 LPSD method

The Logarithmic frequency axis Power Spectral Density ([1], [3]) is an extention of the WOSA tailored for very long time series. While the WOSA uses uniform frequency spacing with the same resolution for every Fourier frequency, LPSD spaces the frequencies approximately uniformly in $\log(f)$ and adjusts the frequency resolution for every Fourier frequency. The main steps are:

1. Establish a set of frequencies at which the PSD will be computed, spaced equally on a logarithmic axis ranging from the minimum to maximum frequency.

2. Determine suitable frequency resolutions for each frequency.

3. For each frequency resolution, compute the corresponding segment length, apply WOSA using that segment length, and report the mean PSD estimate at the required Fourier frequencies.

By varying the segment length as a function of frequency, shorter segments can be used at higher frequencies where small frequency resolution is not needed, allowing one to reduce the variance on the PSD estimate by averaging the PSDs of many short segments.

2.3 logPSD Algorithm

The logPSD algorithm ([1]) is similar to LPSD, but discards the first few DFT bins to reduce the deleterious effects at low frequencies. As a result, longer time segments are needed for low frequencies. The main steps are:

1. Determine the lowest frequency and apply a discard factor to set the length of the first data segment.

2. Use this initial segment length for the first few frequency bins, choosing frequencies as integer multiples of the base frequency.

3. Once the constant-length region is complete, gradually reduce the segment length so that successive frequencies remain evenly spaced on a log scale, with the segment length decreasing as a function of frequency.

4. For each segment length, apply the WOSA method, and calculate the mean PSD estimate at the required Fourier frequencies.

Figure 1 shows the PSD estimates of a time series using the three methods. The differences arise from the choice of frequency resolution. A shorter segment length results in a broader frequency resolution, which tend to smooth out features in the PSD. These effects are more visible in regions where the true PSD changes rapidly. When comparing a measured PSD to a mission requirement, the PSD requirement should be computed using the same frequency resolution as the PSD estimator at that frequency.

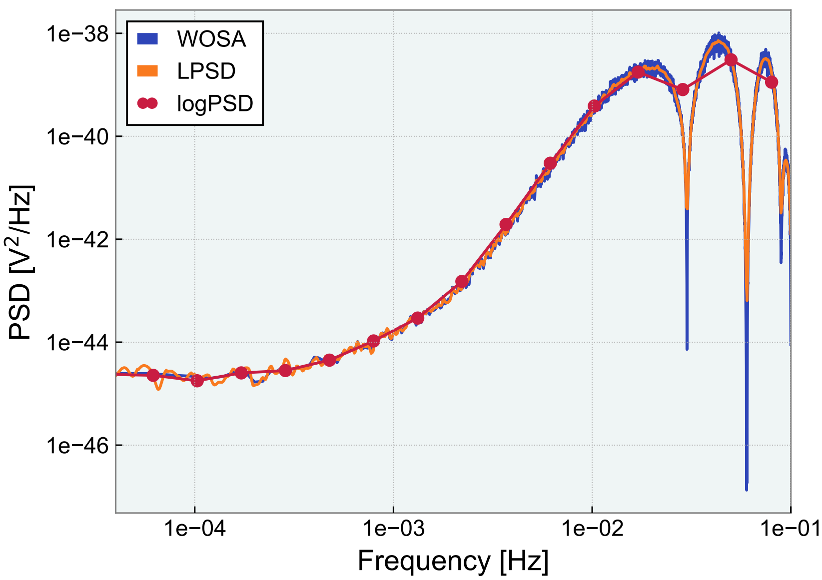

Figure 1: Comparing the three methods for PSD estimation

3 PSD estimation in the presence of transients

In the presence of transient bursts, especially loud ones, the PSD estimate can become significantly distorted. Glitches – sudden and short-lived disturbances in the detectors' data stream – are common sources of such transients. Figure 2 shows that small glitches have little effect on `mean-based` method, while Figure 3 shows that large glitches can distort the PSD.

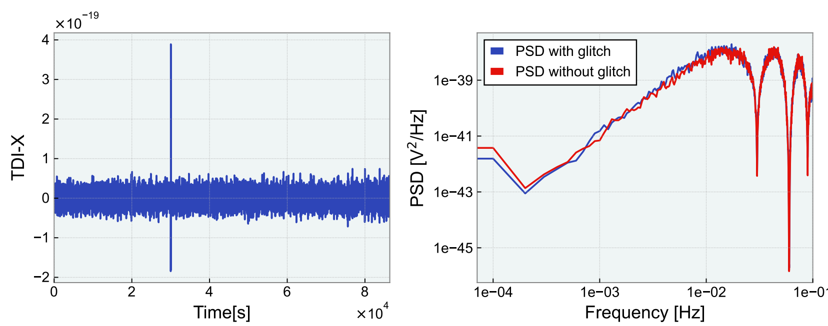

Figure 2: Effect of a small glitch on PSD estimation using `mean-based` method

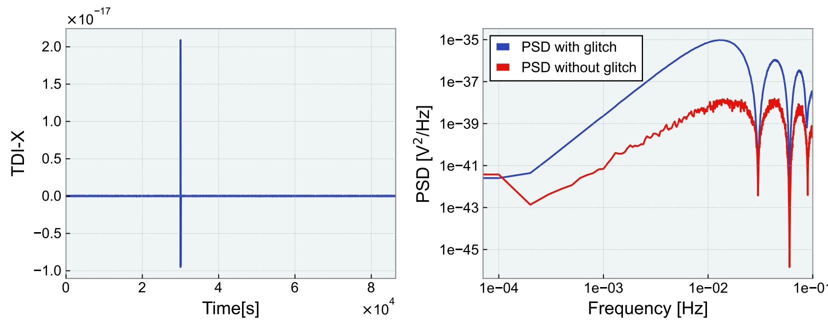

Figure 3: Effect of a larger glitch on PSD estimation using `mean-based` method

4 `Median-based` methods

In each of the `mean-based` methods (WOSA, LPSD, and LogPSD), the final PSD estimate is obtained by averaging the periodograms of each time segment. We propose an alternative: `median-based` methods. Instead of averaging, we take the median of the periodograms. A short transient may have a large impact on the PSD estimate of one or more time segments, resulting in a large impact on the average PSD, but not on the median PSD. The following diagram shows that the `median-based` method is robust to transients.

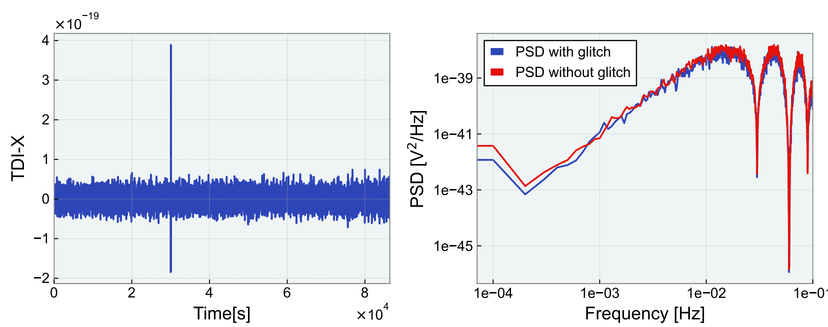

Figure 4: Effect of a small glitch on PSD estimation using `median-based` method

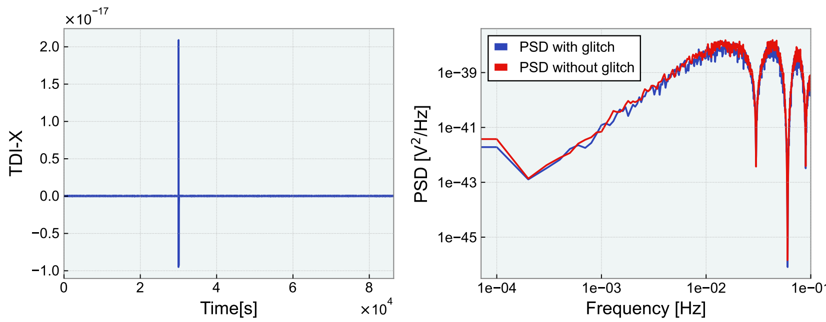

Figure 5: Effect of a larger glitch on PSD estimation using `median-based` method

5 PDF for PSD estimation methods

5.1 PDF for `mean-based` method

In the `mean-based` method, given N segments and a true PSD value s at some given frequency, the observed mean x is expected to follow the chi-squared distribution with $2N$ degrees of freedom (Appendix A):

$$f_{\text{mean}}(x) \propto \left(\frac{x}{s}\right)^{N-1} e^{-\frac{(x/s)}{2}}. \tag{1}$$

5.2 PDF for `median-based` method

In `median-based` method, when N is odd, the measured median m is distributed as:

$$f_{\text{med}}(m) \propto \left(1 - e^{-m/s}\right)^n \left(e^{-m/s}\right)^{n+1}, \quad n = (N-1)/2. \tag{2}$$

This can be derived from the fact that the probability distribution of the PSD estimate x_i for any individual time segment follows a χ^2_2 distribution: $e^{-x_i/s}$. For even N, the distribution is:

$$f_{\text{med}}(m) \propto e^{-2(m/s)} \int_{m/s}^{2(m/s)} dy \left(e^{-y} - e^{-2(m/s)}\right)^{(N/2)-1}. \tag{3}$$

The derivations for equations 2, 3 are outlined in Appendix A. To verify the PDF, we begin by simulating a LISA time series and computing its PSD using a frequency resolution Δf_1, denoted as $\text{PSD}^{\text{true}}(f)$. Next, we generate $n = 1000$ independent time-domain noise realizations produced using this same PSD. For each realization, we compute its PSD using a frequency resolution Δf_2, resulting in:

$$\text{PSD}_1^{\text{noise}}(f),\ \text{PSD}_2^{\text{noise}}(f),\ \ldots,\ \text{PSD}_n^{\text{noise}}(f), \quad n = 1000.$$

We then fix a frequency $f = f_0$, and extract the PSD value at this frequency from both the original and all noise realizations:

$$\mathrm{PSD}^{\mathrm{true}}(f_0), \quad \mathrm{PSD}_1^{\mathrm{noise}}(f_0), \quad \ldots, \quad \mathrm{PSD}_n^{\mathrm{noise}}(f_0), \quad n = 1000.$$

We then construct a histogram of these noise values and compare it to our theoretical PDF, centered at the histogram's mean. Figures 6, 7 confirms the PDF is correct for `mean-based` and `median-based` method.

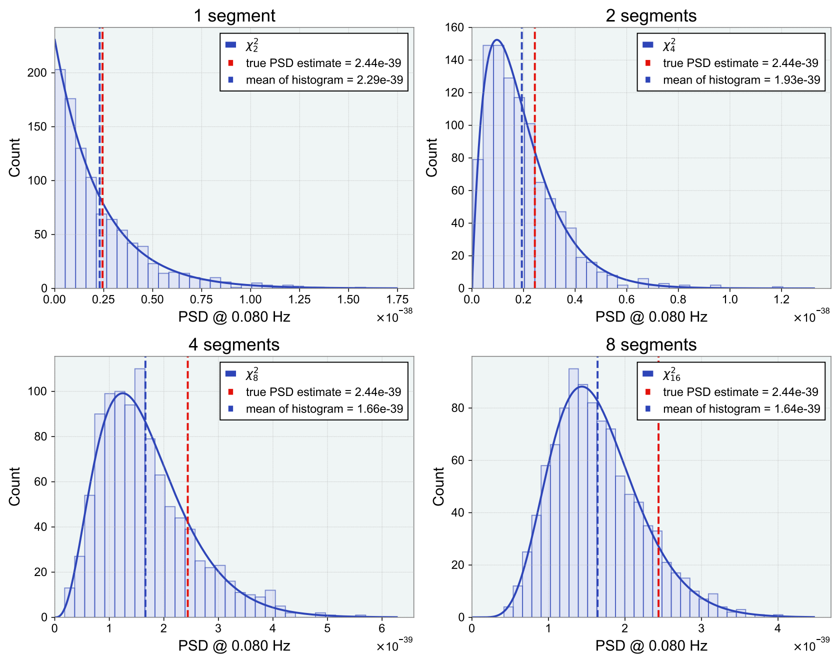

Figure 6: The distribution of PSD estimates for the `mean-based` method at a chosen frequency, along with the theoretically expected PDF.

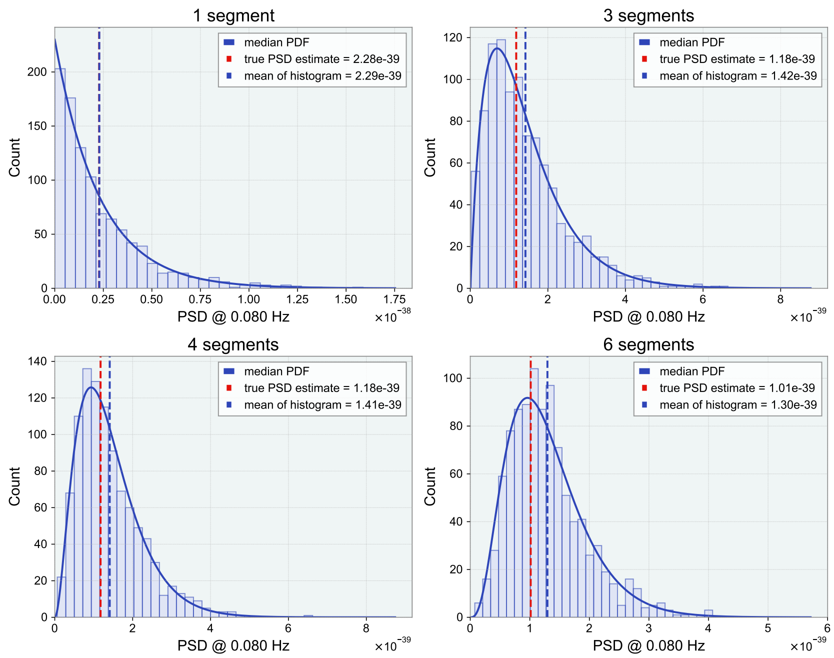

Figure 7: The distribution of PSD estimates for the `median-based` method at a chosen frequency, along with the theoretically expected PDF.

6 Bias in PSD estimate

6.1 Bias due to different frequency resolutions

In Figures 6, 7, the average estimate from the noise realization simulations differs from the true PSD value when the number of segments differs from 1.

$$\mathrm{PSD}^{\mathrm{true}}(f_0) \neq \frac{1}{n}\left(\sum_{i=1}^{n} \mathrm{PSD}_i^{\mathrm{noise}}(f_0)\right), \quad n = 1000. \tag{4}$$

This is due to the effects of smearing due to frequency resolution, as discussed at the end of Section 2.3. When the same frequency resolution is used to calculated the original PSD and the PSDs of noise realizations, the PSDs are equal within expected statistical uncertainties, and the bias is consistent with zero. When $\Delta f_2 > \Delta f_1$ (which results from dividing the time series into more, shorter segments), the mismatch leads to larger bias. As the gap between Δf_1 and Δf_2 grows, the discrepancy between the histogram mean and the true PSD also increases. This difference between the estimated PSD evaluated with one frequency resolution, and the true PSD evaluated at a different frequency resolution, is not in itself a problem, and is intrinsic to PSD estimation. It does however illustrate the importance of matching frequency resolutions when comparing PSD values.

6.2 Inherent bias in `median-based` method

In addition to the bias introduced by the differences in frequency resolution, `median-based` PSD estimation introduces another form of bias, one that stems from using the median instead of the mean. Recall that the time series is divided into N segments, and a modified periodogram is computed for each. The `mean-based` method averages these periodograms, while the `median-based` method takes their median. Since the mean and median are fundamentally different statistical measures, they won't be equal. The `mean-based` PSD estimators are more commonly used and well understood, and in the absence of bias induced by frequency resolution the `mean-based` method is unbiased. In contrast, the `median-based` estimate is biased relative to this reference,

$$\mathrm{PSD}^{\mathrm{true}}(f) = \mathrm{PSD}^{\mathrm{mean}}(f) \neq \mathrm{PSD}^{\mathrm{median}}(f).$$

This difference is also reflected in PDFs of the two methods. As illustrated in Figure 8, when the true PSD is equal to 1, the `mean-based` PDF is centered around 1, while the `median-based` PDF is not, for $N > 1$.

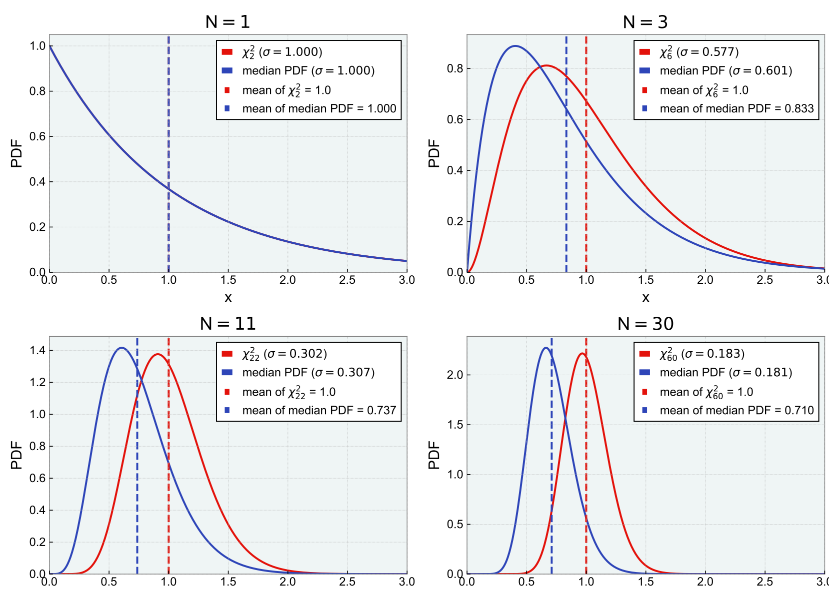

Figure 8: Comparing the PDFs for mean- and `median-based` PSD estimates when the true PSD is 1. The mean PDF is centered at 1, whereas the median PDF is biased.

This bias arises because the PSD estimates for individual time segments follow an exponential distribution $e^{(-x/s)}$, which has a mean of s and a median of $s(\ln 2)$. As a result, the average median of several draws from this distribution will tend to be smaller than the true value s. This bias causes the median method to systematically underestimate the true PSD. To correct for this, we apply a scale factor that aligns the average of the `median-based` PDF with that of the `mean-based` PDF. This scale factor depends only on the number of segments N. Figure 9 demonstrates how applying this factor rescales the median distribution to match the mean distribution's average.

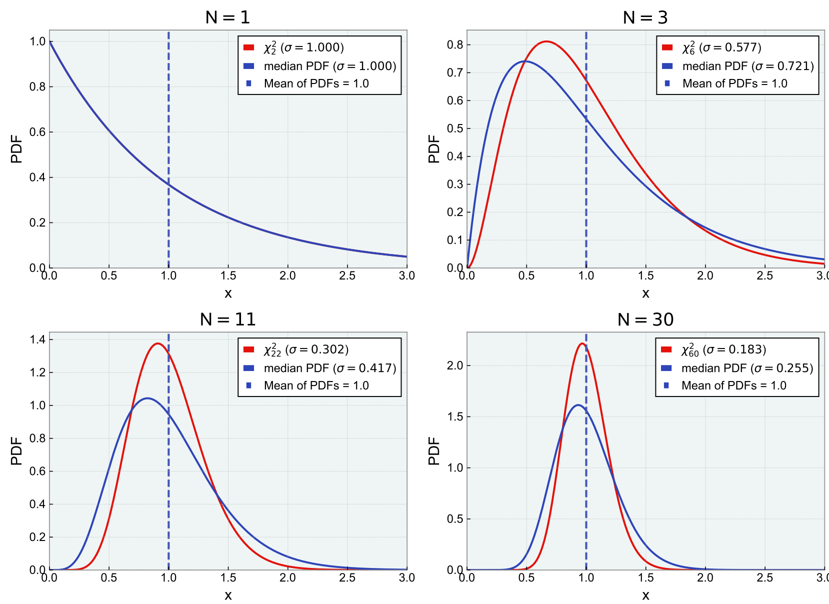

Figure 9: Comparing the PDFs for `mean-based` and `median-based` PSD estimates when the true PSD is 1. The median curve is scaled to have the same average as the mean curve.

Scaling factor: Let $f_{\text{med}}(m)$ denote the PDF of the unscaled `median-based` estimator when the true PSD is s, with N segments. Then the scale factor $g(N)$ is defined such that

$$\int_0^\infty m\, f_{\text{med}}(m)\, dm = s \cdot g(N). \tag{5}$$

This gives us:

$$g(N) = \frac{\int_0^\infty m\, f_{\text{med}}(m)\, dm}{\int_0^\infty x\, f_{\text{mean}}(x)\, dx}. \tag{6}$$

Accordingly, at each frequency we must adjust the median-based estimator by dividing the raw median by $g(N)$, where N may be different at different frequencies (for the LPSD and LogPSD methods at any rate). The scaled `median-based` method's PDF is given by:

$$f_{\text{med}}(x) = \frac{1}{g(N)} f_{\text{med}}\left(\frac{m}{g(N)}\right) \tag{7}$$

7 Uncertainty in PSD estimate

To quantify the uncertainty at a $c\%$ confidence level, we compute the lower and upper confidence limits by taking the $\left(\frac{100-c}{2}\right)$th and $\left(\frac{100+c}{2}\right)$th percentiles of the appropriate PDF centered on the PSD estimate. This can be done by taking the `median-based` method's PDF from Equation 2 or 3 , scaling the x axis by $1/g(N)$ as described in the previous section, and then integrating to get the required percentiles:

8 Limitations of the `median-based` method

Median distributions are roughly 20–40% wider than `mean-based` distributions, as shown in Figure 9 – a statistical trade-off for increased robustness to transients. Achieving the same standard deviation for the median as for the mean would require $30-100\%$ more data.

Furthermore, our method is not robust if a transient appears in half or more of the segments. If a transient appears in half of the segments or more, the median method is not robust, as the middle-most segment will be affected too. Figure 10 illustrates this.

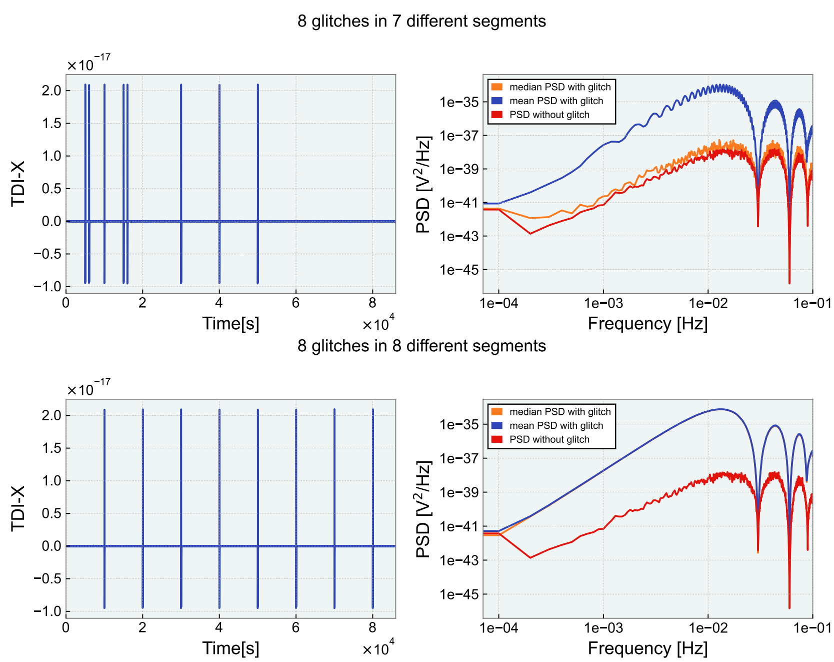

Figure 10: WOSA-median PSD estimate using 16 segments. When half of the segments are affected, the estimate is not robust anymore. Note that in the bottom plot the blue line is hidden under the green line.

This problem can be lessened by reducing the segment length, though this would lead to worse frequency resolution. Finally, the position of transient in the segment matters too. Transients near the edges have less impact due to tapering effect of window function

9 Conclusion

In this report, we examined `mean-based` method (WOSA, LPSD, LogPSD) [1] and showed that they are not robust against large transients. To address this, we introduced `median-based` variants that replace the mean with the median to improve robustness. We also derived the probability distributions for both mean- and `median-based` method and discussed the limitations of the median approach.

References

[1] LISA Project Team ESA, *Power Spectral Density (PSD) Recipe*, Technical Note ESA-LISA-EST-MIS-TN-0004, Issue 1/1, 25 March 2024.

[2] P. D. Welch, *The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms*, IEEE Transactions on Audio and Electroacoustics, vol. 15, no. 2, pp. 70–73, June 1967. https://ieeexplore.ieee.org/document/1161901

[3] Michael Tröbs and Gerhard Heinzel, *Improved spectrum estimation from digitized time series on a logarithmic frequency axis*, Measurement, Volume 39, Issue 2, 2006, Pages 120–129. ISSN: 0263-2241. https://doi.org/10.1016/j.measurement.2005.10.010.

A PDF for `mean-based` method

We prove that the average of PSD estimates from independent time segments follows a χ^2 distribution.

Definition *Let $Z_1, Z_2, \ldots, Z_d$ be a set of i.i.d. standard normally distributed random variables. The sum of squares*

$$\chi_d^2 = Z_1^2 + Z_2^2 + \cdots + Z_d^2$$

is said to have a chi-squared distribution with d degrees of freedom.

Theorem. Let $X_i \sim \chi^2(r_i)$, for $i = 1, 2, \ldots, n$, be independent chi-squared random variables. Then the sum

$$Y = X_1 + X_2 + \cdots + X_n$$

follows a chi-squared distribution with $r_1 + r_2 + \cdots + r_n$ degrees of freedom:

$$Y \sim \chi^2(r_1 + r_2 + \cdots + r_n).$$

For any fixed bin, the real and imaginary parts of the DFT coefficient are normally distributed and i.i.d:

$$\mathrm{Re}\{A_k\},\ \mathrm{Im}\{A_k\} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\big(0, \sigma^2\big)$$

The normalized magnitude-squared follows a chi-squared distribution with 2 degrees of freedom,

$$\frac{|A_k|^2}{\sigma^2} \;=\; \frac{\mathrm{Re}^2 + \mathrm{Im}^2}{\sigma^2} \;\sim\; \chi^2_2.$$

At a given frequency, the mean-based PSD estimate, obtained by dividing the time series into N segments and evaluating at a chosen frequency, is given by

$$\widehat{P} = \frac{L}{NU} \sum_{n=1}^{N} \big|A_n\big|^2. \tag{8}$$

Here, L is the number of points in each segment, $W(j)$ is the window function, $U = \frac{1}{L} \sum_{j=0}^{L-1} W^2(j)$ is the sum of window power, and A_n is the DFT of the n-the windowed segment at the chosen frequency. The PSD estimate is the sum of N random variables, each with 2 degrees of freedom. Hence, $\widehat{P}$ follows chi squared distribution with $2N$ degrees of freedom.

B PDF for `median-based` method

We derive the PDF for the `median-based` method.
For an odd sample size $N = 2n + 1$ of points drawn from a continuous distribution with cumulative distribution function (CDF) $F(v)$ and PDF $f(v)$, the event "median $= m$" occurs when:

$$n \text{ samples} < m, \quad 1 \text{ sample} = m, \quad n \text{ samples} > m.$$

These outcomes follow a trinomial distribution with probabilities:

$$p_1 = F(m), \quad p_2 = f(m)\, dm, \quad p_3 = 1 - F(m).$$

Hence we have:

$$f_{\mathrm{med}}(m)\, dm = \frac{(2n+1)!}{n!\, 1!\, n!} \left[F(m)\right]^n \left[f(m)\, dm\right]^1 [1 - F(m)]^n, \quad n = \frac{N-1}{2}. \tag{9}$$

This gives us the PDF:

$$f_{\mathrm{med}}(m) \propto F(m)^n \left[1 - F(m)\right]^n f(m), \quad n = \frac{N-1}{2}. \tag{10}$$

For $f(x) = \frac{1}{s}e^{-x/s}$ and $F(x) = 1 - e^{-x/s}$,

$$f_{\text{med}}(m) \propto \left(1 - e^{-m/s}\right)^n \left(e^{-m/s}\right)^{n+1}, \quad n = \frac{N-1}{2}. \tag{11}$$

For the even sample size $N = 2n$, let the samples be $X_{(1)}, X_{(2)}, \ldots, X_{(2n)}$, where $X_{(k)}$ is k-th order statistic. The joint PDF of $X_{(n)}$ and $X_{(n+1)}$ is:

$$f_{\text{med}}(X_{(n)}, X_{(n+1)}) \propto [F(X_{(n)})]^{n-1} f(X_{(n)}) f(X_{(n+1)}) [1 - F(X_{(n+1)})]^{n-1}. \tag{12}$$

Define the sample median as

$$m = \frac{X_{(n)} + X_{(n+1)}}{2}.$$

Do the following change of variables: $X_{(n)} = y$, and $X_{(n+1)} = 2m - y$ to get:

$$f_{\text{med}}(y, m) \propto [F(y)]^{n-1} f(y) f(2m - y) [1 - F(2m - y)]^{n-1}. \tag{13}$$

To get $f_{\text{med}}(m)$, we integrate over all the y, and substitute the PDF and CDF for the exponential distribution:

$$f_{\text{med}}(m) \propto e^{-2(m/s)} \int_{m/s}^{2(m/s)} dy \left(e^{-y} - e^{-2(m/s)}\right)^{(N/2)-1}. \tag{14}$$